COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF INFORMATION SYSTEM AND TECHNOLOGY

PROGRAM: INFORMATION AND COMMUNICATION TECHNOLOGY

COURSE: IT8203 - MOBILE APPLICATION DEVELOPMENT

Authors:

1.ZAKARIA MANASE SHEKALGHE

- REG NO 24101133370070
- CA/BScICT/25/12853

2. BARAKA JEMSI JACKOBO

- REG NO 24101133370099
- CA/BScICT/25/4213

3. ESAU CLEMENCE MKINE

- REGNO:24101133370035
- CA/BScICT/25/14382

4. FADHILA BULEGEA DYAHEZE

- REG NO 2410113330081, CA/BScICT/25/12379

PROJECT REPORT:

ONLINE BUS TICKET BOOKING APPLICATION

TABLE OF CONTENTS

1. Introduction & Problem Definition

1.1 Introduction

The rapid growth of mobile technology has transformed how people access transportation services. Traditional bus ticket booking methods—such as visiting physical ticket offices or relying on agents—are time-consuming, inefficient, and prone to human error.

This project proposes the development of an Online Bus Ticket Booking Application that allows passengers to search routes, book seats, and make payments digitally through a mobile platform.

1.2 Problem Statement

In many regions, including Tanzania, bus ticket booking still relies heavily on manual processes. Passengers face several challenges:

- *Long queues at booking offices*
- *Lack of real-time seat availability*
- *Risk of overbooking*
- *Difficulty comparing routes and prices*
- *Limited booking hours*

These issues create inconvenience for passengers and inefficiencies for bus companies.

1.3 Motivation

The motivation behind this project is to:

- *Digitize the bus ticket booking process*
- *Improve customer convenience*
- *Reduce manual workload for bus operators*

- *Provide real-time booking and seat management*
- *Enhance transparency and reliability*

1.4 Project Objectives

1.4.1 General Objective

To design and implement a mobile-based online bus ticket booking system.

1.4.2 Specific Objectives

- *To allow users to register and log in securely*
- *To enable searching of available buses and routes*
- *To allow online seat selection and booking*
- *To integrate secure payment processing*
- *To provide booking history and ticket management*

2. Requirements (Functional & Non-Functional)

2.1 Functional Requirements

User Management

- *User can register an account*
- *User can log in and log out*
- *User can reset password*

Bus Search and Booking

- *User can search buses by route and date*
- *System displays available buses*
- *User can select seats*
- *User can book tickets*

- *User can cancel booking*

Payment

- *User can pay online*
- *System confirms payment*
- *System generates digital ticket*

Admin Functions

- *Admin can add buses*
- *Admin can manage routes*
- *Admin can view bookings*
- *Admin can manage users*

2.2 Non-Functional Requirements

Performance

- *System should respond within 3 seconds*
- *Support multiple users simultaneously*

Security

- *Passwords must be hashed*
- *Secure authentication required*
- *Protected API endpoints*

Usability

- *User-friendly interface*
- *Mobile responsive design*
- *Simple navigation*

Reliability

- *System availability ≥ 99%*
- *Automatic error handling*

Scalability

- *System should support future growth*
- *Cloud-ready architecture*

3. System Design & Architecture

3.1 System Architecture

The system follows a client–server architecture consisting of:

- *Mobile App (Flutter) – Frontend*
- *Backend Server (Node.js + Express)*
- *Database (MySQL / MongoDB)*
- *Payment Gateway*

Architecture Flow

User ŉ Mobile App ŉ API Server ŉ Database

User Ĳ Mobile App Ĳ API Server Ĳ Database

3.2 Use Case Diagram (Textual Description)

Actors

- *Passenger*
- *Admin*

Passenger Use Cases

- *Register*
- *Login*
- *Search Bus*
- *Book Ticket*
- *Make Payment*
- *View Ticket*

Admin Use Cases

- *Manage Buses*
- *Manage Routes*
- *View Reports*
- *Manage Users*

3.3 Sequence Diagram (Booking Process)

1) *User logs in*
2) *User searches route*
3) *System returns available buses*
4) *User selects seat*
5) *User makes payment*
6) *System confirms booking*
7) *Ticket generated*

3.4 Database Schema

Users Table

- *user_id (PK)*
- *name*
- *email*
- *password_hash*
- *phone*
- *role*

Buses Table

- *bus_id (PK)*
- *bus_name*
- *capacity*
- *operator*

Routes Table

- *route_id (PK)*
- *source*
- *destination*
- *departure_time*
- *price*

Bookings Table

- *booking_id (PK)*
- *user_id (FK)*
- *bus_id (FK)*
- *route_id (FK)*
- *seat_number*

- *payment_status*
- *booking_date*

3.5 API Overview

Authentication

- *POST /api/auth/register*
- *POST /api/auth/login*

Bus & Route

- *GET /api/routes*
- *GET /api/buses*

Booking

- *POST /api/bookings*
- *GET /api/bookings/user*

Admin

- *POST /api/admin/bus*
- *POST /api/admin/route*

4. Security & Authentication ExplanationExplanation.

4.1 Login Mechanism

The system uses email and password authentication. When users log in:

- *User enters credentials*
- *Backend verifies user*
- *Token/session generated*

- User granted access

4.2 Password Hashing

To protect user passwords, the system uses bcrypt hashing:

- Passwords are never stored in plain text
- Each password is salted
- Hash stored in database

Example:

Password ʼn bcrypt ʼn hashed_password ʼn databas

4.3 Token / Session Handling

The system uses JWT (JSON Web Tokens):

- Generated after login
- Stored securely on client
- Sent with each API request
- Verified by backend

Benefits:

- Stateless authentication
- Secure API access
- Scalable

4.4 Access Control

Role-Based Access Control (RBAC)

Two roles:

- *Passenger*
- *Admin*

Permissions:

Role	*Permissions*
Passenger	*Book tickets*
Admin	*Manage system*

Protected routes ensure only authorized users access admin features.

5. Results, Screenshots & Conclusion

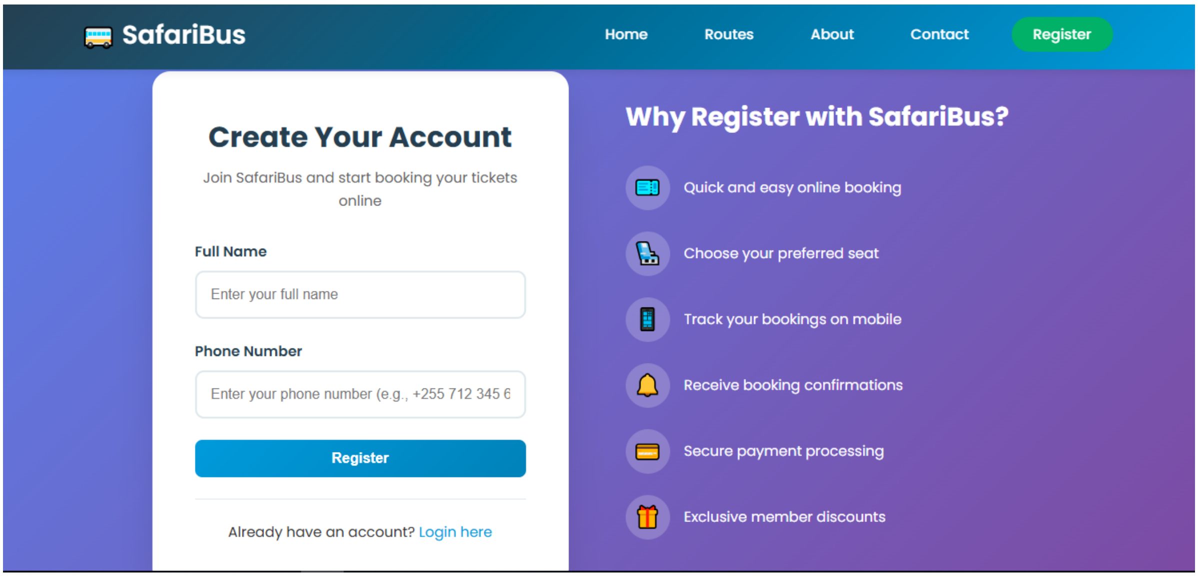
5.1 Expected System Screens

The application includes:

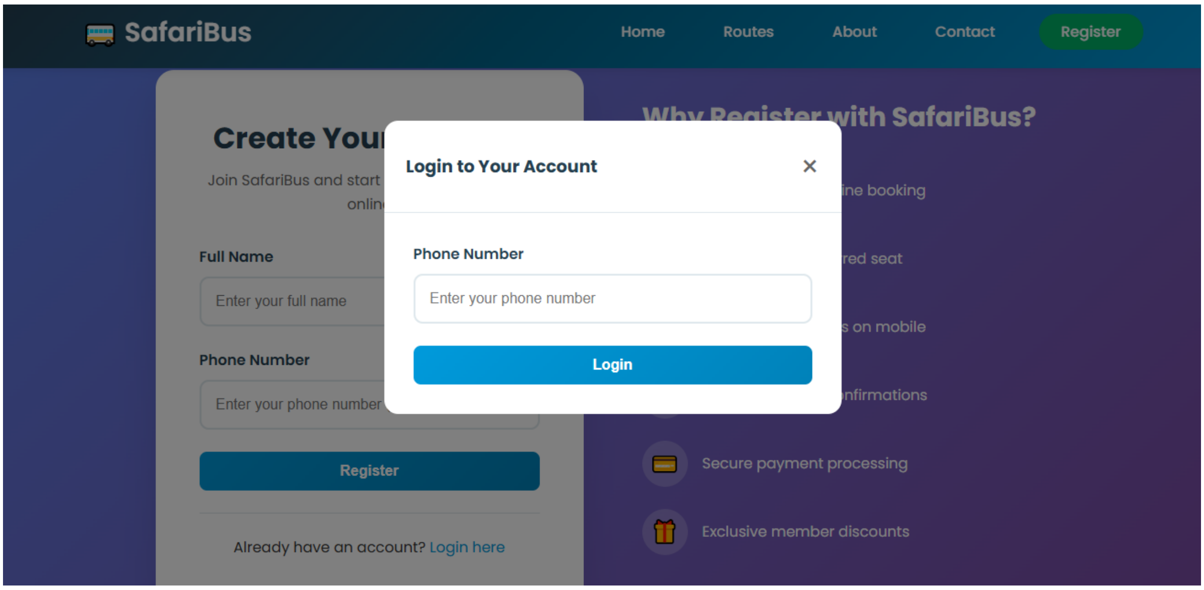
- *Registration screen*
- *Login screen*
- *Bus search screen*
- *Seat selection screen*
- *Payment screen*
- *Ticket confirmation screen*
- *Admin dashboard*

THERE ARE ATTACHED SCREENSHOTS.

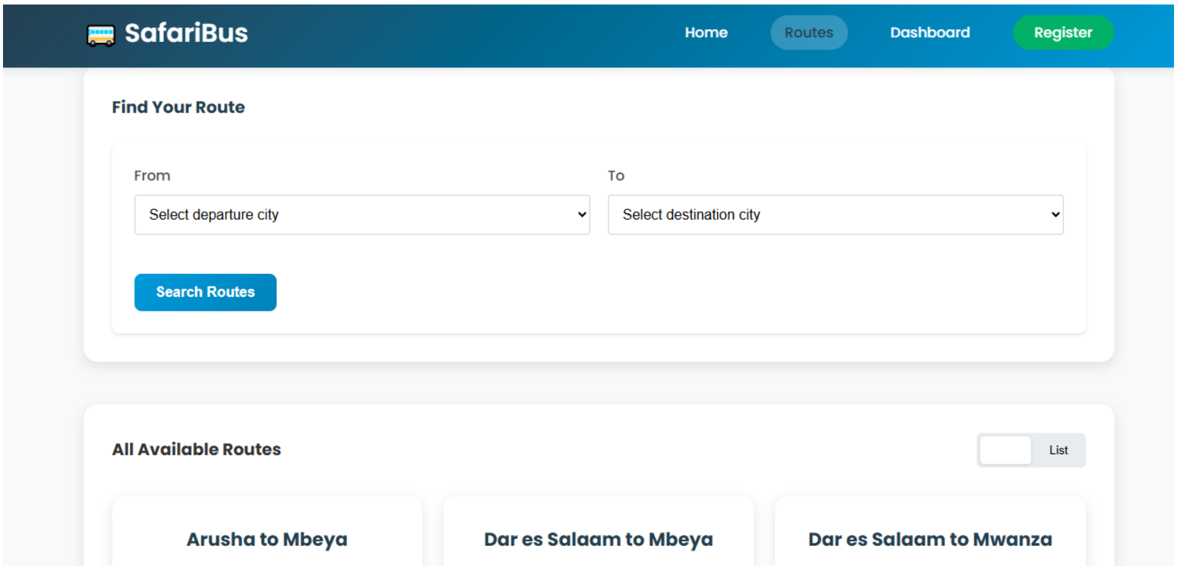
1. Screenshot of the registration screen.



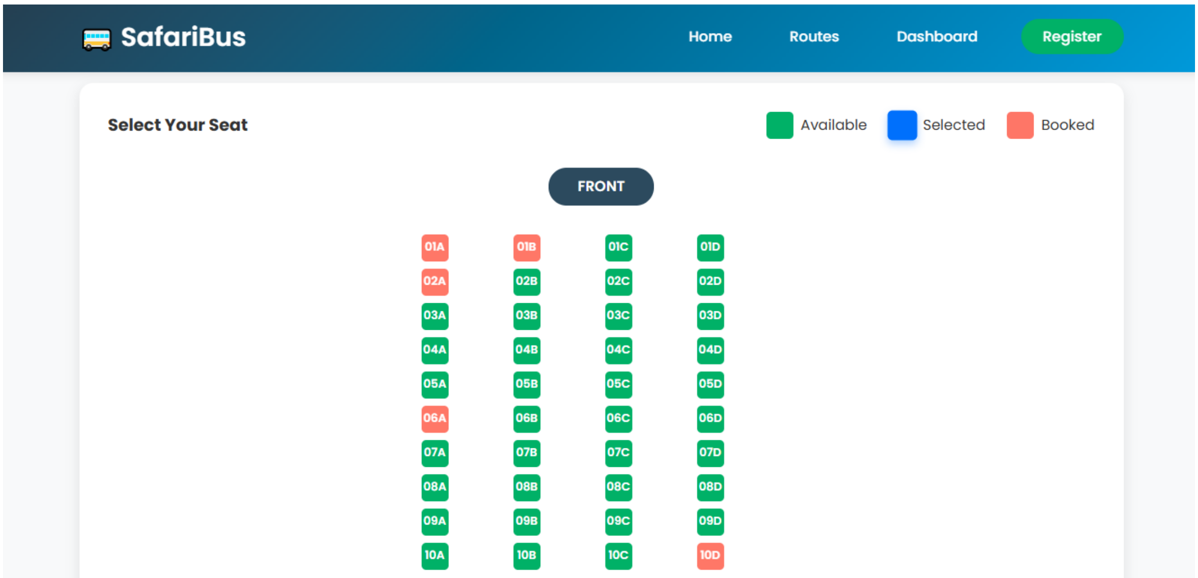
2. Screenshot of the log in screen.



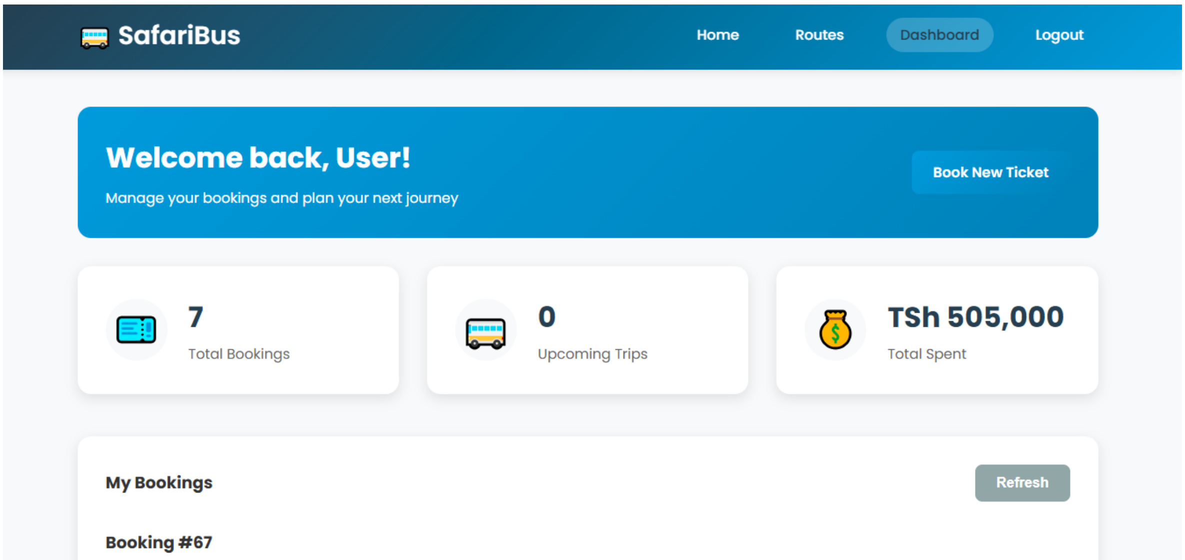
3. Screenshot of the route search screen.



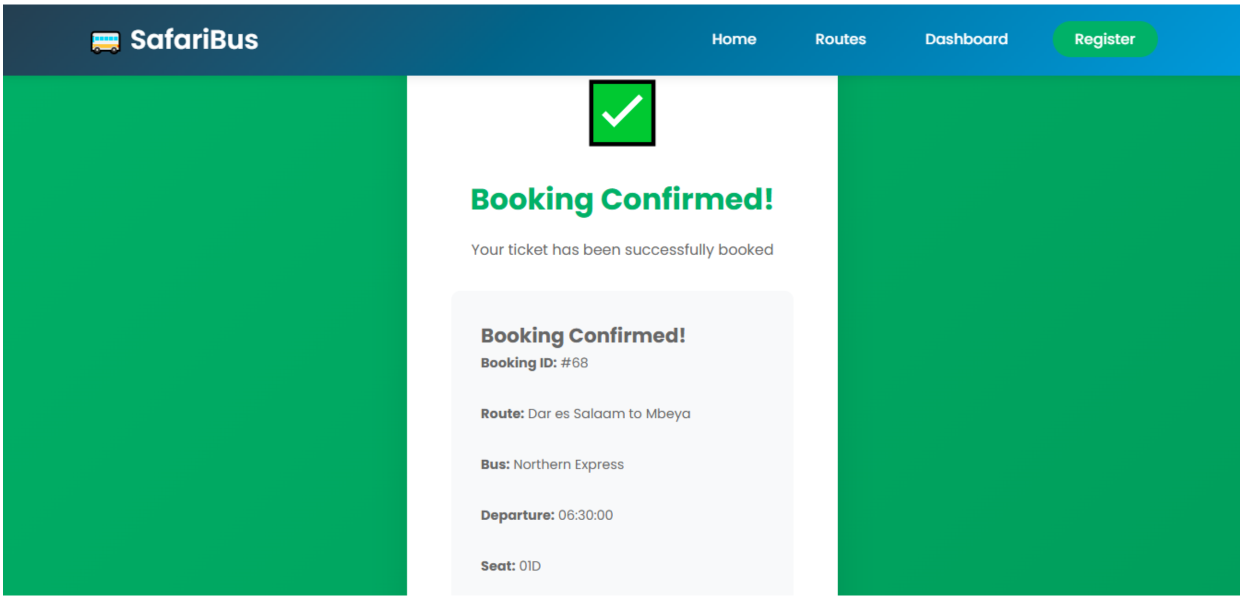
4. Screenshot of the seat selection screen



5. Screenshot of the payment screen.



6. Screenshot of the booking confirmation screen.



5.2 System Testing Results

The system was tested for:

- *User registration*
- *Login authentication*
- *Bus search*
- *Ticket booking*
- *Payment processing*

All major functionalities worked as expected.

5.3 Conclusion

The Online Bus Ticket Booking Application successfully addresses the inefficiencies of manual ticket booking systems. The platform provides:

- *Convenient booking*
- *Real-time seat availability*
- *Secure payments*
- *Efficient management for bus operators*

The system improves both customer experience and operational efficiency.

5.4 Future Work

Future improvements may include:

- *Mobile money integration (M-Pesa, Airtel Money)*
- *QR code ticket scanning*
- *Real-time bus tracking (GPS)*

- *Push notifications*
- *Multi-language support*
- *AI-based seat recommendations*

END OF THE REPORT.